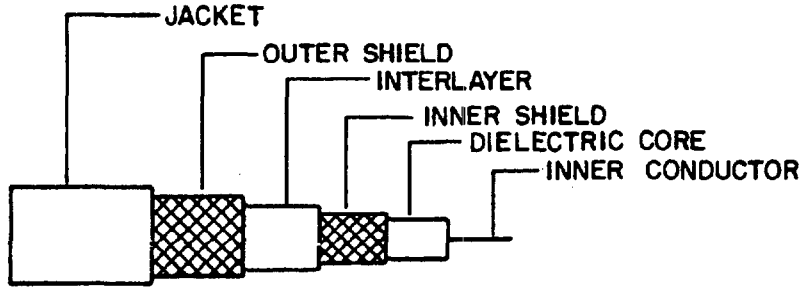
MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, TRIAXIAL, 50-OHM, WATER BLOCKED AND NON-WATER BLOCKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-17.



CABLE SHOULD BE CONSTRUCTED OF NON-TOXIC MATERIAL

FIGURE 1. Triaxial cable, radio frequency M17/135-00001, -00003 (water blocked).

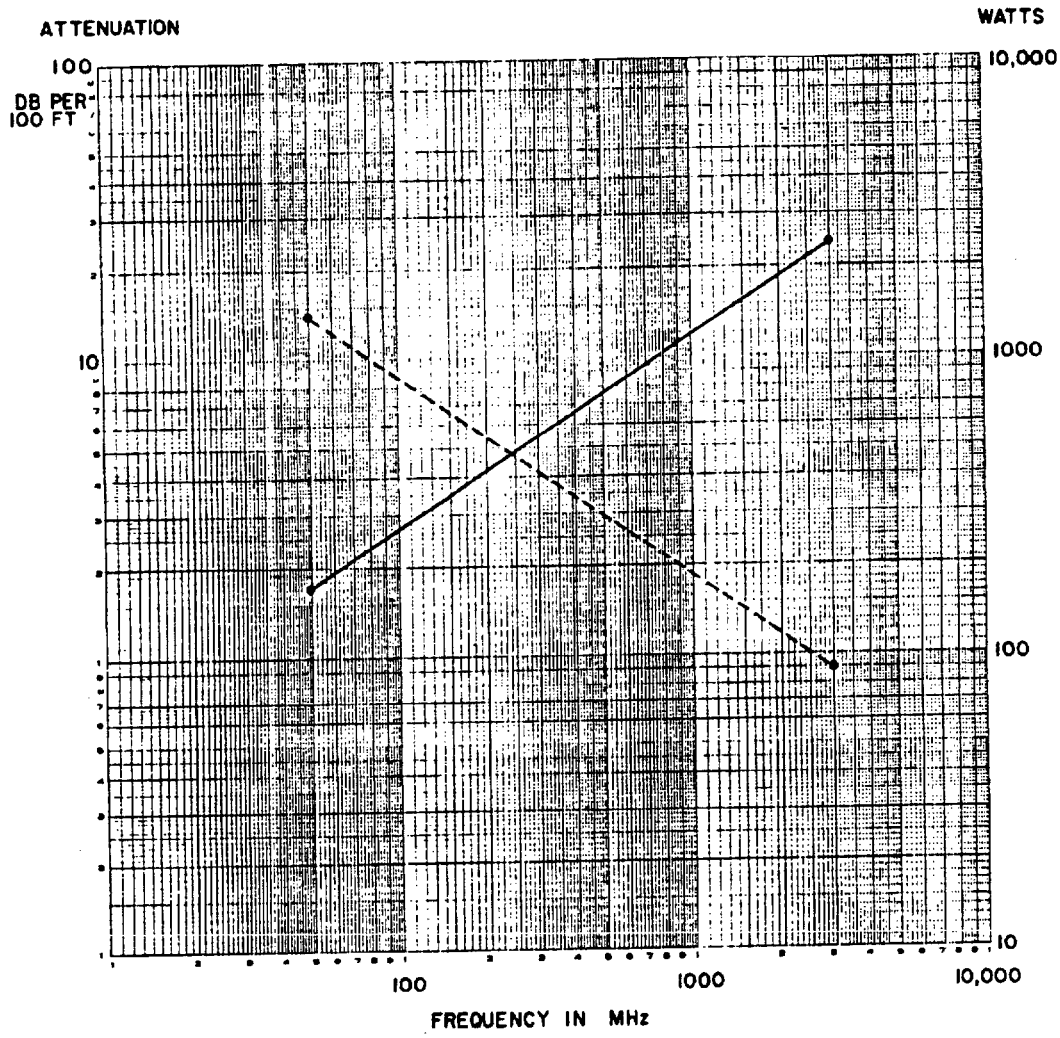
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TABLE I. Description.

Components	Construction
Inner conductor	Seven strands of silver-coated copper wire each strand 0.0296 inch diameter Diameter: 0.089 inch ±0.001
Inner conductor M17/135-00003 M17/135-00004	Solid silver-coated copper wire Diameter: 0.081 inch ±0.001
Dielectric core	Type A-1: Solid polyethylene Diameter: 0.285 inch ±0.010
Inner shield	Single braid of AWG no. 33 silver-coated copper wire Diameter: 0.340 inch maximum Coverage: 90% minimum Carriers: 24 Ends: 7 Picks/inch: 8 Separator tape optional
Interlayer	Type A-1: Solid polyethylene Diameter: 0.365 inch ±0.010
Outer shield	Single braid AWG no. 33 silver-coated copper wire Diameter: 0.414 inch maximum Coverage: 85% minimum Carriers: 24 Ends: 8 Picks/inch: 5.11 Separator tape optional
Jacket M17/135-00001 1/ M17/135-00002	Type XII: Polyurethane Diameter: 0.500 inch ±0.010
Jacket M17/135-00003 1/ M17/135-00004	Type IIIa: Polyethylene Diameter: 0.500 inch ±0.010

1/ Free strippable blocking compound needed where necessary to meet hydrostatic test. The inner shield may be partially imbedded in the dielectric. Applicable to M17/135-00001 and -00003 only.

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Maximum attenuation ————

Maximum power ---- at 25°C sea level

NOTE: (Test requirements shall be noted as line indicated on graph).

Frequency MHz	Attenuation dB	Watts
50	1.7	1400
100	2.6	850
400	6.4	350
1000	11.4	190
3000	23	90

FIGURE 2. Power rating and attenuation.

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SWR	Reflection coefficient	Return loss dB	SWR	Reflection coefficient	Return loss dB
1.3767	.1585	16	17.3910	.8913	1
1.3290	.1413	17	8.7242	.7943	2
1.2880	.1259	18	5.8480	.7079	3
1.2528	.1122	19	4.4194	.6310	4
1.2222	.1000	20	3.5698	.5623	5
1.1957	.0891	21	3.0095	.5012	6
1.1726	.0794	22	2.6146	.4467	7
1.1524	.0708	23	2.3229	.3981	8
1.1347	.0631	24	2.0999	.3548	9
1.1192	.0562	25	1.9250	.3162	10
1.1055	.0501	26	1.7849	.2818	11
1.0935	.0447	27	1.6709	.2512	12
1.0829	.0398	28	1.5769	.2239	13
1.0736	.0355	29	1.4985	.1995	14
1.0653	.0316	30	1.4326	.1778	15

Frequency MHz	Min SRL
50	27.5
1000	21
2000	16
3000	13.8

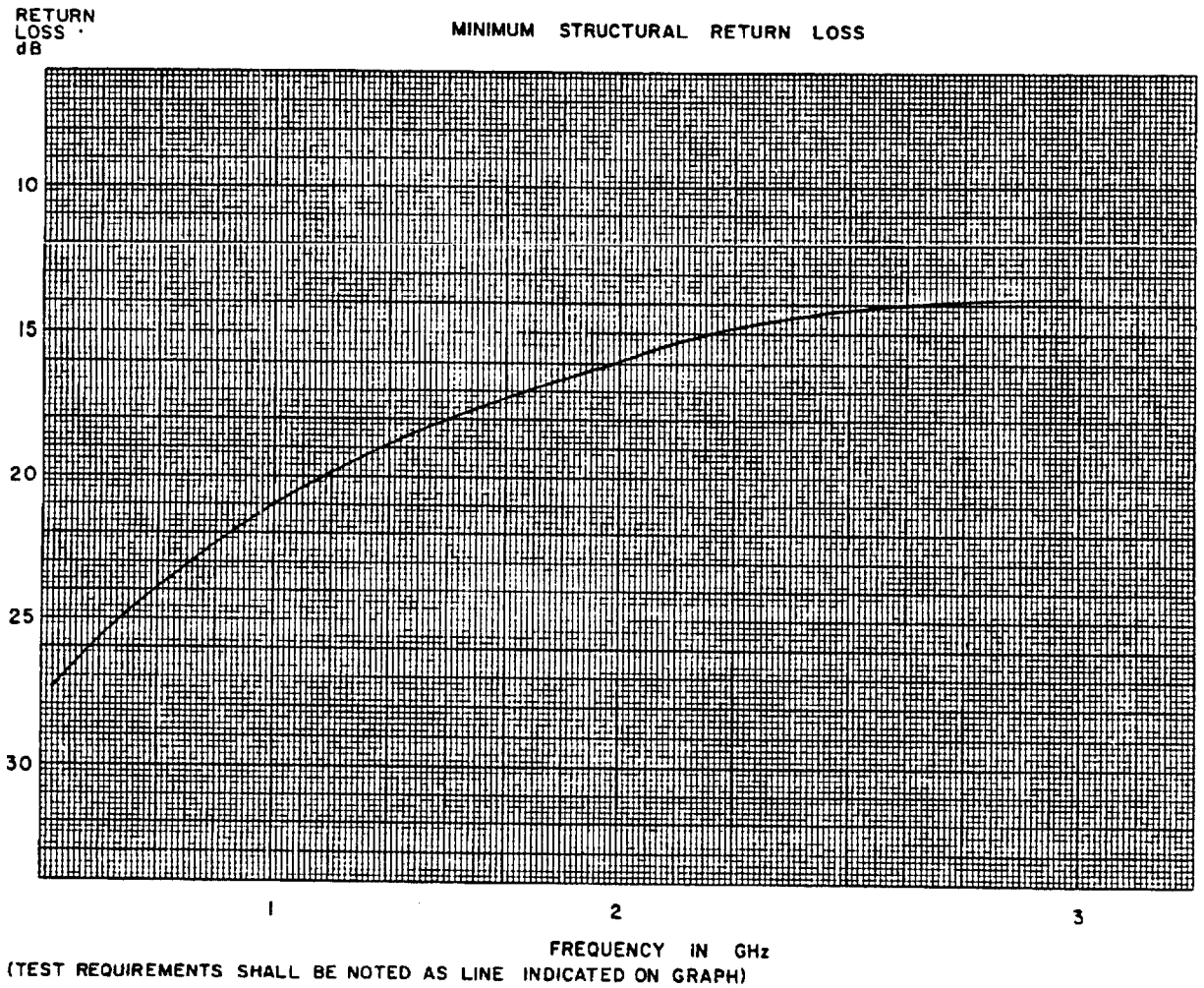


FIGURE 3. Structural return loss.

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ENGINEERING INFORMATION:

Continuous working voltage: 3,700 V rms, maximum.

Operating frequency: 3 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +70°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 0.162 ohm per 100 feet. (0.173 ohm per 100 feet for M17/135-00001 and -00002).

Elongation: 25 percent, minimum; 30 percent, minimum for M17/135-00001 and -00002.

Tensile strength: Not applicable.

Engineering notes: This cable useful for nonhosing low temperature applications.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

M17/135-00001 and -00003 7 pounds, minimum, 65 pounds, maximum.
M17/135-00002 and -00004 7 pounds, minimum, 50 pounds, maximum.

Aging stability: +90°C ±2°C (+98°C ±2°C for M17/135-00001 and -00002).

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend: -55°C ±2°C.

Dimensional stability:

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Not applicable.

Flame propagation: Not applicable.

Acid gas generation: Not applicable.

Halogen content: Not applicable.

Immersion tests: Not applicable.

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Smoke index: Not applicable.

Toxicity index: Not applicable.

Durometer hardness: Not applicable.

Weathering: Not applicable.

Abrasion resistance: Not applicable.

Tear strength: Not applicable.

Heat distortion: Not applicable.

Physicals (aged): Not applicable.

Tensile strength and elongation: Not applicable.

Shrinkage: Not applicable.

Jacket flaws: Not applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 18.5 pounds per 100 feet, maximum (16 pounds per 100 feet, maximum for M17/135-00001 and -00002).

Watertightness (Not applicable to M17/135-00002, 00004): One end of a 5-foot length of completed cable shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing gland in the terminal fitting shall be a metal alloy having a maximum melting point of +88°C. The specimen shall be subjected to a water pressure of 25 lb_f/in^2 for a period of 6 hours. Any evidence of water leakage shall be a cause for rejection.

Hydrostatic (open end) (Not applicable to M17/135-00002, 00004): A 5-foot specimen of completed cable shall be fitted into a gland similar to USL 44500B-45. The cable shall be capable of withstanding 1,000 lb_f/in^2 applied to the gland end of the specimen for a period of 2 hours. Any evidence of water leakage shall be cause for rejection.

Electrical:

Spark test: 8,000 V rms, +10 percent, -0 percent.

Voltage withstanding: 10,000 V rms, minimum, applied between the inner and outer conductors with the outer conductor grounded, 500 V dc applied between the outer conductor and the extra shield with the extra shield grounded.

Insulation resistance: Not applicable.

Corona extinction voltage: 5,000 V rms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: See figure 2.

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Structural return loss: See figure 3.

Capacitance: 32 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

Contamination: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation	Type
M17/135-00001 1/	AA-3833	Water blocked
M17/135-00002 1/	TRF-8	Non-water blocked
M17/135-00003	---	Water blocked
M17/135-00004	---	Non-water blocked

1/ Inactive for new design.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 17, 85, 99
DLA - IS
NASA - NA

User activities:
Army - AV
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1094)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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